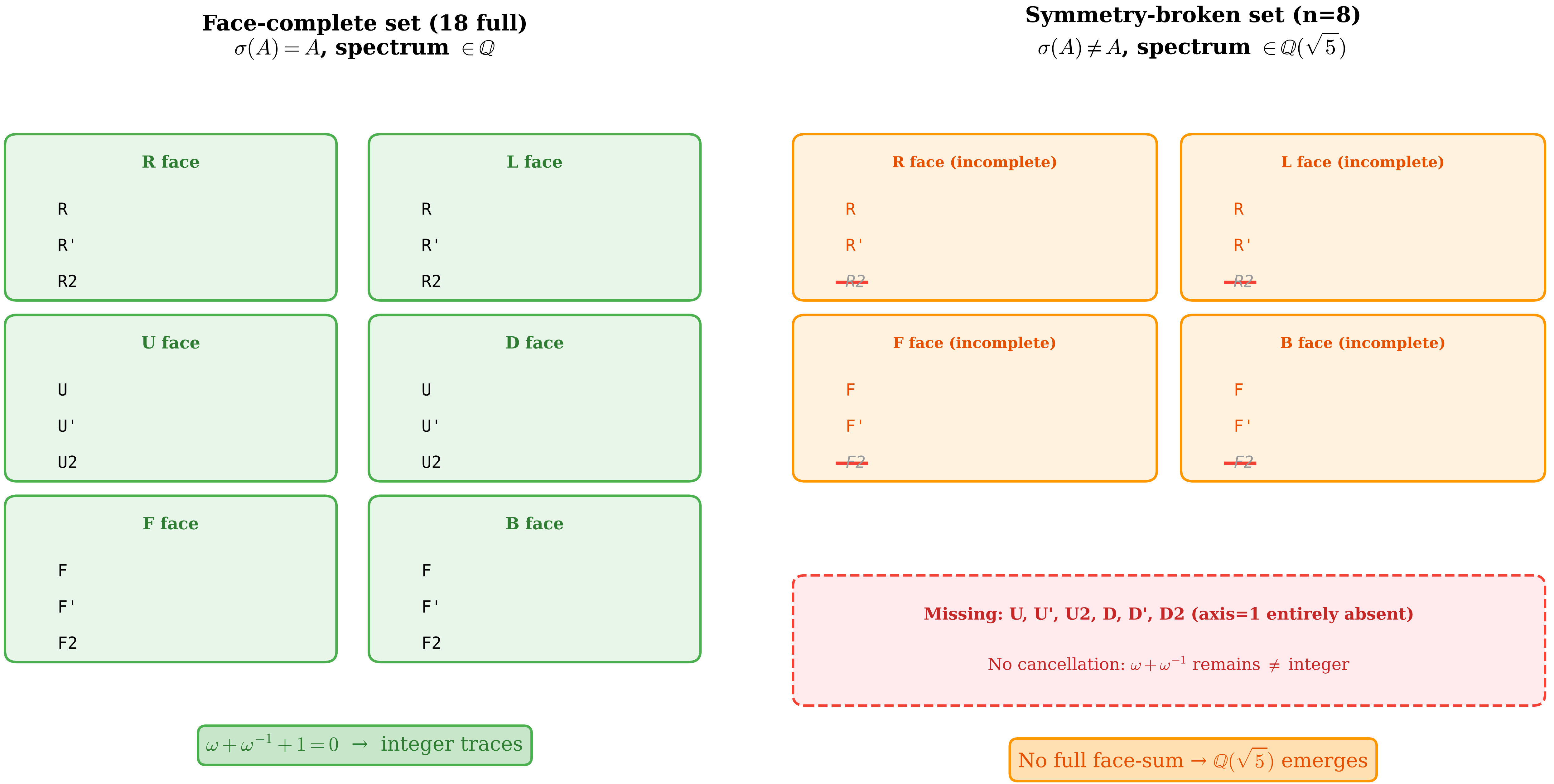
Face-complete set (18 full)
$\sigma(A) = A$, spectrum $\in \mathbb{Q}$
R face
R
R'
R2
L face
R
R'
R2
U face
U
U'
U2
D face
U
U'
U2
F face
F
F'
F2
B face
F
F'
F2
$\omega + \omega^{-1} + 1 = 0$ → integer traces
Symmetry-broken set (n=8)
$\sigma(A) \neq A$, spectrum $\in \mathbb{Q}(\sqrt{5})$
R face (incomplete)
R
R'
~~R2~~
L face (incomplete)
R
R'
~~R2~~
F face (incomplete)
F
F'
~~F2~~
B face (incomplete)
F
F'
~~F2~~
Missing: U, U', U2, D, D', D2 (axis=1 entirely absent)
No cancellation: $\omega + \omega^{-1}$ remains $\neq$ integer
No full face-sum → $\mathbb{Q}(\sqrt{5})$ emerges